Boundedness

A real sequence is an ordered list a_1, a_2, a_3, … indexed by the positive integers. Its order and repetitions matter. The set of values taken by a sequence does not retain all this information.

A sequence is bounded above if there is a real number U such that $a_n \le U$ for every n. It is bounded below if there is a real number L such that $L \le a_n$ for every n. It is bounded if both conditions hold. Equivalently, there is $M \ge 0$ with $|a_n| \le M$ for every n.

The sequence $a_n = 1/n$ is bounded: $0 < a_n \le 1$ for every positive integer n. The sequence $b_n = (-1)^n$ is also bounded, because $-1 \le b_n \le 1$. Boundedness describes where the terms can lie; it does not say that they approach a single value.

A bound need not be a term of the sequence, and it need not be the best possible bound. For $a_n = 1/n$, both 1 and 2 are upper bounds, but 1 is the least upper bound of the set of values. The lower bound 0 is never attained.

Do not confuse “all terms remain in a fixed finite interval” with “the terms eventually stay close to one point.” The first describes boundedness; the second belongs to convergence.

Questions to explain
What does a bound have to control? Can a bounded sequence keep oscillating? Must a lower bound be one of the terms?

Convergence and quantifiers

A sequence a_n converges to a real number ℓ if, for every $\varepsilon > 0$, there exists a positive integer N such that, for every $n \geq N$, $|a_n - \ell| < \varepsilon$.

The order of the choices matters. N may depend on ε, but after N has been selected it must work for every later index n. The definition does not allow a new N to be chosen separately for each term.

For $a_n = 1/n$, given $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. For every $n \geq N$, $0 < 1/n \leq 1/N < \varepsilon$. Thus the sequence converges to 0.

Every convergent real sequence is bounded. To see this, use $\varepsilon = 1$ to place all sufficiently late terms in the interval $(\ell - 1, \ell + 1)$. There are only finitely many earlier terms. A sufficiently large bound can cover those terms as well as the tail.

The converse is false. The bounded sequence $b_n = (-1)^n$ alternates between -1 and 1 and does not converge. If it converged to ℓ, then for $\varepsilon = 1/2$ all sufficiently late terms would lie within $1/2$ of ℓ. Taking a late even index and a late odd index would give $2 = |1 - (-1)| \leq |1 - \ell| + |\ell + 1| < 1$, a contradiction.

Questions to explain

Why must the same N work for all later terms? Why do finitely many early terms not prevent convergence or boundedness? Why is boundedness insufficient for convergence?

Monotone convergence

A sequence is nondecreasing if $a_{n+1} \geq a_n$ for every n. It is nonincreasing if $a_{n+1} \leq a_n$ for every n. A sequence is monotone if it has either of these properties. Equal consecutive terms are allowed.

Every bounded monotone sequence of real numbers converges.

More precisely, a nondecreasing sequence that is bounded above converges to the supremum of its set of values. A nonincreasing sequence that is bounded below converges to the infimum of its set of values.

Proof in the nondecreasing case: Let s be the supremum of $\{a_n : n \geq 1\}$, which exists by completeness. Given $\varepsilon > 0$, $s - \varepsilon$ cannot be an upper bound for all the terms. Therefore some term a_N satisfies $s - \varepsilon < a_N \leq s$. By monotonicity, for every $n \geq N$ we have $s - \varepsilon < a_N \leq a_n \leq s$. Hence $|a_n - s| < \varepsilon$, proving convergence to s.

Both ideas matter. Boundedness supplies a finite supremum; monotonicity makes the entire tail stay above the term already found near that supremum. Boundedness alone permits oscillation. Monotonicity alone permits divergence to infinity, as with $a_n = n$.

Questions to explain

Where is completeness used? Where is monotonicity used? Why does finding just one term near s not by itself control every later term?

Subsequences

A subsequence of a_n is a sequence a_{n_k} obtained by choosing a strictly increasing sequence of positive integer indices $n_1 < n_2 < n_3 < \dots$. The indices must keep moving forward, but the values need not be increasing.

If a_n converges to ℓ, every subsequence also converges to ℓ. Given $\varepsilon > 0$, choose N for the original sequence. Strictly increasing positive integer indices satisfy $n_k \geq k$. Therefore, when $k \geq N$, the term a_{n_k} is also far enough along the original sequence to be within ε of ℓ.

A convergent subsequence does not imply that the whole sequence converges. For $b_n = (-1)^n$, the even-indexed subsequence is constantly 1 and the odd-indexed subsequence is constantly -1. Each subsequence converges, but the original sequence does not.

The Bolzano–Weierstrass theorem for real sequences states that every bounded real sequence has a convergent subsequence. It guarantees a convergent subsequence, not convergence of the entire sequence.

Questions to explain

Why must subsequence indices be strictly increasing? How do the even and odd subsequences distinguish boundedness from convergence? Which word changes the claim in Bolzano–Weierstrass?